



Web accessibility of healthcare Web sites of Korean government and public agencies: a user test for persons with visual impairment

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Abstract

The present study attempts to evaluate the accessibility of government and public agency healthcare Web sites in Korea through user testing. The study recruited blind (24 participants) and second-level sight-impaired people (1 participant), and evaluated ten healthcare Web sites of Korean government and public institutions. These Web sites revealed problems across four principles of accessibility, i.e., perceivable, operable, understandable, and robust. First, according to principle 1, perceivable, the main errors occurred in providing adequate alternate texts and clear instructions. Second, five errors were predominant with regard to principle 2, operable. These are keyboard operation errors, focus order errors, bypass blocks errors, page title errors, and the lack of appropriate link text. Third, with respect to principle 3, understandable, most problems were found in user responses, content sequence, tables, and labels. Finally, based on principle 4, robust, problems regarded markup correctness and Web application accessibility. The problems identified in the study can be simply resolved; however, similar accessibility issues have recurred. Therefore, future research needs to investigate whether government and public agencies or their Web developers are aware of the importance of accessibility.

Keywords Web accessibility · Healthcare · Government Web sites · Visual impairment · WCAG 2.0 · South Korea

1 Introduction

Recently reported infectious diseases, such as the Zika and MRSA viruses, and the radioactive contamination caused by the nuclear accident in Fukushima, Japan, have been notable threats to public health. As such horrific events continue to occur, people have seriously recognized the need for reliable information to respond to these threats properly. In general, people tend to search for multiple sources when the quality of information is difficult to judge [1]. Among the various sources of health information, the government and public agencies are perceived as having a higher authority than other sources [2]. In the era of Government 3.0, online health information provided by government and public institutions needs to be actively open to the public and shared.

Owing to the development of multimedia information sources, online health information is provided not only in

a text-oriented fashion, but also in a variety of audiovisual formats. This property is one of the salient phenomena of the digital age, through which many people can quickly and easily search for health information on the Web, as well as make health-related decisions (e.g., prevention and treatment); thus, appropriate Web accessibility must be ensured for the purposes of seeking health information. Web accessibility allows people with physical disabilities—such as those related to sight, hearing and movement—to be free to use the information on the Web in ways comparable to non-disabled users [3]. Web accessibility technically supports them to contribute to the Web through interaction with it [4]. However, people with disabilities or the elderly have been omitted from the benefits of using online health information. According to the information gap survey conducted in Korea in 2012, most of the general public uses the Internet, whereas only 56% of people with disabilities use the Internet in Korea [5].

Lately, Korea has begun to pay attention to Web accessibility. Legal grounds include the Anti-Discrimination Act for Disabled People enacted in April 2007, the Act on the Welfare of People with Disabilities Partially Revised in 2008, and the National Informatization Act amended in May 2009,

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all of which stipulate the obligations of Web accessibility. The Web accessibility quality mark certification system has been introduced and implemented since 2007 [6]. Korea established the Korean Web Content Accessibility Guideline (KWCAG) 1.0 in December 2005 with reference to the international standard WCAG 1.0. In December 2010, Korea developed KWCAG 2.0 as the national standard with reference to WCAG 2.0. Like WCAG 2.0, KWCAG 2.0 consists of four principles: perceivable, operable, understandable, and robust/technologically advanced. Under the main principles, KWCAG 2.0 has 13 guidelines and 22 detailed inspection items [7]. In 2004, Korea developed KADO-WAH 1.0, a Web accessibility evaluation program, and then KADO-WAH 2.0 (2006), KADO-WAH 3.0 (2009), and K-WAH 4.0 (2012). K-WAH 4.0 is a tool to automatically test compliance with the six detailed guidelines based on the KWCAG 2.0. The six detailed guidelines are (1) providing appropriate alternative text; (2) providing an appropriate title; (3) specifying default language; (4) opening new windows as a response to a user's request; (5) providing labels, and (6) preventing markup errors [8].

As the population ages, the use of the Internet by the elderly increases. Considering these phenomena, Korea needs to ensure Web accessibility that meets international standards. Although there have been many studies on Web accessibility in Korea, few studies have examined the accessibility of health-related government and public agencies' Web sites. Previously, the researcher evaluated the accessibility of these Web sites with automated testing and expert testing [9]. As a follow-up to the previous study, the present study attempts to evaluate the accessibility of government and public agency healthcare Web sites through user testing, by which the study aims to perform a comprehensive evaluation complementing the automated or expert testing.

2 Previous studies on user testing and related works

Previous studies that performed user testing according to WCAG addressed similar problems regarding Web accessibility [10–15]. People with disabilities have reported that navigation is difficult due to the illogical and complicated structure of Web pages, such as page layouts that contain excessive information or that require too many steps to work on the Web sites. In addition to the inappropriate structure, mouse-over menus and pop-up windows made it difficult for users to navigate the Web sites because assistive technologies were not compatible with them; thus, they needed to handle all tasks with the keyboard only. Next, inconveniences occurred when the size of elements, such as buttons and drop-down lists, as well as text, was too small, when the number of links to be connected was too many or when Web

page headings were ambiguous. Regarding the interaction between a user and the system, there were code errors in checking whether the system's response to the user's work was inconsistent or not.

Other challenging issues included the lack of alternative means, such as text, for multimedia contents or the difficulty of locating the cursor or an input field when the entire Web page could not be seen. Above all, users experienced considerable barriers from Web applications; for example, CAPTCHA, which displays random characters to prevent spam programs and test users, was an unlabeled graphic, and AJAX, a technology that updates only parts of a Web page without having to refresh all of it hindered users' smooth Web access by making it difficult for users to scan pages or to access certain functions and elements using assistive technologies.

Previous research on Web accessibility performed user testing for diverse types of Web sites. User testing on job-seeking sites in the U.S. uncovered numerous accessibility problems [10]. These problems included confusing layouts and links, incompatibility with JAWS, ambiguous descriptions and labels, unlabeled form controls, lack of clear input fields, flash content errors, lack of skip navigation, illogical tab order, etc. Web accessibility assessments of electronic voting prototypes in Norway have revealed problems such as sequence errors, inappropriate Web page structure, navigation issues, and the lack of codes to check for vote submission [11]. User testing on various Web sites in the UK, including government, the private sector, the public sector, and nonprofit and commercial organizations is reported in [12]. The study identified such issues as illogical content sequences, inadequate functionality, the lack of responses on users' performance, and incorrect headings. Another user testing on the Web site of Thailand Cyber University noted the lack of alternative text for graphics and audio and video content, considerable navigation errors, and the demand for excessive scrolling [13]. Users with visual impairment evaluated Blackboard, a type of learning management system in [14]. The users identified focus errors and critical user response problems. Table 1 summarizes accessibility problems identified in the previous user testing.

Regarding the accessibility of government and public agencies' Web sites, Jager [16, 17] investigated the level of compliance of e-government Web sites with a policy presented in Sect. 508 of the US Rehabilitation Act. The study employed multi-method evaluations to evaluate accessibility in a comprehensive way. For automated testing, the study used a free tool to quantitatively measure ten Web sites' rates of compliance with Section 508. Next, the expert testing involved using an instrument that consisted of scale questions with possible scores from 0 to 20 and open-questions, derived from the Section 508 regulations. The participants for the user testing were

Table 1 Accessibility problems identified in literature

Illogical page structure
Poorly organized and cluttered page layouts
Levels of navigation too complicated
Mouse-over menus difficult to navigate
Pop-up windows difficult to navigate
Elements (font, buttons, drop-down lists, etc.) too small
Redundant links
Inappropriate link text
Ambiguous page headings
Inappropriate heading structure
Inconsistent response to user's request
Insufficient feedback to whether the user's action was effective
Lack of alternative text for multimedia contents
Difficult to locate the cursor or input field
Problems with the accessibility of Web applications (CAPTCHA, AJAX, JavaScript, etc.)

people with a visual or mobility impairment, who were asked to perform tasks assigned and answer questions regarding accessibility problems they experienced while performing the tasks. The study [16] indicated that the results of expert testing were similar to the ones of user testing, which were more in-depth and detailed. The automated testing, however, had certain limitations and as a result missed accessibility problems that expert testing or user testing could identify. The Web sites tested had major accessibility problems, such as incompatibility issues with screen readers, screen enlargement, and alternate color schemes, which were serious barriers to offering equal accessibility to everyone.

Brobst [18] conducted a Web content review, automated testing, and expert testing for federal health care Web sites of the US. The Web content review was conducted for 20 Web sites to quantitatively and qualitatively assess policy statements concerning Web accessibility. Based on the same instrument that Jager [16] employed, the study performed expert testing to identify accessibility issues that the automated testing could not check. The findings indicated that only 15 Web sites presented the accessibility statements; among them 17 sites were assessed as high quality and 8 sites as medium. Automated testing indicated that only 33 percent of the Web sites provided appropriate accessibility; 67 percent failed the test. Through expert testing, Brobst addressed the two main issues: one is the lack of alternative tag description, and the other is the difficulty of direct access to embedded interfaces. Furthermore, the study [18] highlighted that federal policies such as the Americans with Disabilities Act of 1990 and Section 508 were inappropriate for promoting Web accessibility.

3 Web accessibility in Korea

The studies that employed user testing based on KWCAG revealed similar problems as follows [8, 19–22, 24–26]. One of the most salient problems was the lack of alternative text. Specifically, there were duplicate alternate texts or an absence of appropriate alternative text. Additionally, there was a problem with the size of the text: the text was too small or a size adjustment function was not available. Other complicated issues were the complexity of the Web site and the difficulty in accessing the information itself. In other words, it is difficult to access information due to complicated installation programs, and people with visual impairments cannot access the password input method (e.g., CAPTCHA) for the security program.

Previous studies investigated the accessibility of homepages in general schools, special education schools, and universities. According to the results of automated testing using the K-WAH 3.0, Kim [19] divided school homepages into two groups: high and low accessibility. The two groups had notable differences in search times. Web sites with high accessibility took less time than those with low accessibility. Students with disabilities had bigger time differences in searching the two sites compared to general students, indicating that compliance with Web accessibility is more effective for users with poor accessibility. The findings of user testing on the Web sites of special education schools addressed problems stemming from low compliance with KWCAG [8]. Common issues included incompatibility with screen enlargement and font size adjustment, and color schemes. In addition, essential information was hidden in the sub-menu of the bulletin board in the form of an attached file. Such a complex structure made it difficult to find what users needed. Likewise, Ahn [20] divided university homepages based on the Web accessibility compliance rate of K-WAH 4.0 and performed user testing on them. According to Ahn [20], the structures of the Web sites could not be read by the screen reader because the sites had too many menus and multiple frames and lacked alternative text. Moreover, users experienced difficulty registering and incompatibility with screen readers.

Jung [21] conducted user testing for the open banking sites of four domestic companies that have improved accessibility for Internet banking sites. The study identified several critical problems. There were focus errors when the screen was updated, while skip navigation errors were inconvenient for users. This study confronted the same issues with security software, which was an extension of the Internet banking problems related to personal firewalls and security processes that were not completely addressed. User testing on sixty-two public library Web

sites in Korea observed that alternative text and frame titles were lacking [22]. Users were unable to access the sites using only screen readers and keyboards; thus, they failed to fulfill such tasks as accessing the site, searching for books, reading e-books, registering for membership, and asking questions regarding information resources.

Regarding the accessibility of Korean government and public agencies Web sites, previous studies evaluated homepages of the central administrative agencies, the National Election Commission, provincial governments and health care institutions. Moon and Moon [23] found that sites that adhere to the guidelines for Web accessibility have increased Web usability for both people with and without disabilities, while sites that do not have reduced Web usability for both groups. Testing of the Web site of the National Election Commission identified similar problems [24]. The screen enlargement function was not effective due to the too small size of the text and button images. Users could not navigate it with a keyboard only because the menu configurations, the layouts and the structures of the Web sites were too complex. Moreover, alternative text was lacking, functional errors occurred (e.g., search function), and many uses of JavaScript were difficult to access; for example, the eBooks could not be controlled by the keyboard. Web sites of provincial governments had several of the same issues as the Web site of the National Election Commission [25]. The Web pages composed of frames, time-dependent JavaScript, and installation of Web applications (e.g., ActiveX) were challenging to attain accessibility. In addition, the violation of the first principle of KWCAG, perceivable, stood out in the Web sites of provincial governments. Problems included the lack of both appropriate alternative text and specific description for the content and inappropriate titles for the Web pages.

Yi [9] performed automated and expert testing of accessibility for health care Web sites of government and public agencies in Korea. The study observed key issues such as inappropriate alternative text and color contrast, errors in the default language display and markup, bypass blocks, and content linearization. Above all, non-compliance with keyboard use hindered people with disabilities from accessing health information. Another study of Korean hospital Web sites' accessibility for people with low vision addressed users' difficulty to understand graph information, location of hospitals, and floor guidance [26]. Table 2 presents accessibility problems identified in previous studies on Korean Web sites.

4 Methods

Not only can user testing reflect the difficulties of people with disabilities, it also offers a more detailed and in-depth analysis. User testing, on the other hand, is labor-intensive

Table 2 Accessibility problems identified in literature (Korea)

Duplicated or lack of alternative text
Font size too small
Enlargement function unavailable
Complicated page structure
Complicated installation programs
Problems with the accessibility of CAPTCHA and JavaScript
Compatibility problems with screen readers
Color scheme hard to read

and time-consuming compared to automated or expert testing [17]. The study conducted user testing according to KWCAG 2.1, which was developed based on WCAG 2.0, and contained 22 test items, as presented in Table 3. Released in March 2015, KWCAG 2.1 outlines how to develop a Web site that allows people with disabilities access to Web content on an equal basis as non-disabled users [27, 28].

The questions for user testing were based on the previous study [17] as follows:

1. What assistive technologies (e.g., screen readers, joysticks, etc.) are you using?
2. What difficulties are there in navigating within this Web site?
3. How difficult is it to read the text on this Web site?
4. How difficult is it to use the search function on this Web site?
5. How difficult is it to use applications (e.g., download files, use video or audio files, fill out forms) on this Web site?
6. Overall, what is the difficulty of using ancillaries on this Web site?
7. Have you found any problems that other people with disabilities may experience?

Vision impairment in Korea has 6 levels. According to the results of the pilot study conducted on five participants, people with vision impairment can be divided into two groups, one consists of those who are blind (level 1) and those who are close to blindness (level 2), and the other is those who were in other visual impairment levels. People with level 3 through 6 could access information on the Web by enlarging the content, whereas people with level 1 and 2 could not. Thus, the two groups reported significantly different findings. People at visual impairment levels 3–6 mainly identified accessibility issues regarding enlargement. People at level 1 and 2, however, addressed comprehensive accessibility problems. Accordingly, the formal study recruited volunteers who belong to only the first group, blind (24 people) and second-level vision-impaired people (1 person). The

Table 3 Evaluation items of KWCAG 2.1

	Evaluation items
Perceivable	
1.1.1 Appropriate alternative text	All non-text content has a text alternative that serves the equivalent purpose
1.2.1 Captions	For multimedia content, captions, manuscripts, or sign language interpretations are provided
1.3.1 Content perception without colors	Color is not used as the only visual means of conveying content
1.3.2 Clarified instructions	Instructions provided do not rely solely on sensory characteristics such as shape, size, visual location, orientation, color, or sound
1.3.3 Text content contrast	The visual presentations of text and images of texts have a contrast ratio of at least 4.5:1
1.3.4 No audio background	No audio on a Web page plays automatically
Operable	
2.1.1 Keyboard operation	All functionality of the content is operable through a keyboard interface
2.1.2 Focus order	Using a keyboard interface, components receive focus in an order that preserves meaning, and the keyboard focus indicator is visible
2.2.1 Time adjustable	For each time limit that is set by the content, the user is allowed to adjust it
2.2.2 Pause control	For scrolling or auto-updating information, a mechanism is provided for the user to pause or control it
2.3.1 Avoiding flashes and blinks	Web pages do not contain anything that flashes or blinks 3–50 times in any one second period
2.4.1 Bypass blocks	A mechanism is available to bypass blocks of content that are repeated on multiple Web pages
2.4.2 Page titled	Web pages, frames, and content have titles that describe the topic or purpose
2.4.3 Appropriate link text	The purpose of each link can be determined from the link text
Understandable	
3.1.1 Language of page	The default human language of each Web page can be programmatically determined
3.2.1 User response	A change of context (opening a new window, moving focus) is not automatically initiated without the user's intention
3.3.1 Content sequence	A correct reading sequence that preserves the meaning of the content can be programmatically determined
3.3.2 Tables	Information and structure conveyed through tables can be programmatically determined or are available in text
3.4.1 Labels	Labels or instructions are provided when content requires user input
3.4.2 Error correction	If an input error is detected, the error is described to the user and suggestions for corrections are provided
Robust	
4.1.1 Markup correctness	In content implemented using markup languages, elements have complete start and end tags, are nested and contain attributes defined according to their specifications
4.2.1 Accessible Web applications	Web applications (JavaScript, Applet, ActiveX, Flash, etc.) within content are accessible to user agents, including assistive technologies, or through alternatives

study participants were composed of 17 males and 8 females in their 20s (1 person), 30s (11 people), 40s (9 people), 50s (3 people), and 60s (1 person). It is notable that the study participants consisted of 11 Web accessibility experts as well as 14 non-experts. Regarding their average Internet use in hours on a weekly basis, expert users spent 39 h while non-experts spent 16 h; thus, on average, the participants accessed information on the Web for 27 h per week.

User testing was conducted at the Korea Federation of the Blind. The Korea Federation of the Blind is not only good for disability users, but also adequately equipped to carry out the present study. For the test Internet Explorer was used as the browser and Sense Reader as the screen reader. The participants listened to the sound with headsets or earbuds.

The study participants were given specific tasks that corresponded to the above questions. The tasks were

modified from the previous study [17], considering the characteristics of the healthcare Web sites in Korea. For instance, many of the Web sites require registration; otherwise, information accessed on the site is restricted. In addition, when registering, a user should go through a verification process with his or her cell phone:

- Register/sign-in
- Visit/browse Web pages. Navigate pages that correspond to the main menu
- Use the search function
- File download and print. Applicable to test Web applications
- Input contact information for user verification/leave a message in the customer service center.

The researcher described the task that the user should perform before beginning each task. When the user did not understand or ask questions during the process, the researcher repeated the tasks out loud again. While performing the user tests, a digital recorder was used to record all of users' comments and questions about accessibility problems that occurred during the tasks; users' comments and questions, and the answers for them were also transcribed. The study questionnaire was filled out during as well as after performing the tasks. The study participants were encouraged to talk about the accessibility problems that they experienced after completing each task and whole tasks once again. The user testing took 2–4 h; each participant received an incentive of approximately \$40.

There are 31 health-related government and public agencies in Korea. Considering time issues for user testing, the present study selected 10 institutions among the 31, as presented in Table 4, referring to the two criteria of the previous study [17]. One was whether the Web site would be a priority concern for people with disabilities. This was particularly to evaluate Web sites that are closely related to these people, such as the Korea Disabled People Development Institute and the National Rehabilitation Center. The other was the degree to which the information provided by the organization was of interest to the public in relation to health issues. This criterion was to identify the troubles that people with disabilities had while accessing these major health-related Web sites.

5 Results and discussion

Most of the study participants noted similar problems across all health-related Web sites observed in the present study. Table 5 presents the major accessibility problems identified for each Web site. Interestingly, the accessibility experts among the participants identified potential errors that non-experts did not find and suggested specific solutions to fix them. Accessibility problems caused some users to fail at

task completion; some users gave up a task as soon as they found it hard to perform, for example, when there was no alternative text for an image, while some users passed a task after trying over 20 min. The study did not include the failed cases when measuring the average time of each of task completion. The majority of users, however, completed the task by repeatedly attempting or getting additional help from research assistances, even if it took time.

5.1 Results

First, the study conducted user testing for the National Cancer Center (NCC) Web site. NCC is a national organization that performs professional treatment and research for cancer [27, 28]. Regarding the page navigation task, none of the users failed. It took them about 1–8 min, 4 min on average to complete the task. As the key issues, users noted the lack of appropriate alternative text and content linearization errors. For example, at the top of the list page, there should be a 'View' or 'Search' button right beside the search box, though it only displayed 'New Window Open'; thus, users were not sure what to open. Also, when moving with a tab instead of a direction key, the tab went directly to the appointment time in the doctor's name, skipping the appointment date; thus, users could not set up the appointment date. Regarding the search function, users indicated errors of appropriate link text and alternative text. Since there was no specific description other than 'more links' or 'new window', users were confused about what the link was about or what information would be presented in the new window. In addition, due to the absence of alternative text, a user could not tell whether the search results were displayed in the order of 'most recent' or 'most popular'.

For the fourth task, download, none of the users failed. It took them about 30 s to 6 min, 3 min on average to complete the task. Users reported focus order errors and the lack of proper link text. Regarding focus order errors, users mentioned that when a dialog box opened, the download list (including save/open option) should be automatically accessible through the tab key, though on this site the save/open link button was difficult to access and errors occurred in the dialog box because the tab key did not work. For the proper link text problems, users found that the text should display 'National Cancer Center annual report download link' although it read 'National Cancer Center annual graphic link', thus, it was likely to be misunderstood as just an image rather than a download link, as presented in Fig. 1.

Second, the study tested the National Health Insurance Service (NHIS) Web site. NHIS manages all citizens' health insurance enrollment, fee collection, and medical fee schedules [29]. In NHIS, the first task was to register/login on the site: One user failed, and it took about 2 to 5 and a half minutes, 3 and half minutes on average, to complete the task.

Table 4 Web sites for user testing

1. National Cancer Center (http://www.ncc.re.kr)
2. National Health Insurance Corporation (http://www.nhis.or.kr)
3. National Health Information Portal (http://health.mw.go.kr)
4. Ministry of Health and Welfare (http://www.mw.go.kr)
5. Korea Center for Disease Control and Prevention (http://www.cdc.go.kr)
6. National Medical Center (http://www.nmc.or.kr)
7. National Rehabilitation Center (http://www.nrc.go.kr)
8. Public Health Portal (http://www.g-health.kr)
9. Ministry of Food and Drug Safety (http://www.mfds.go.kr)
10. National Center for Mental Health (http://www.ncmh.go.kr)

Table 5 Key accessibility issues identified in user testing

Web sites	Key accessibility problems identified
National Cancer Center (NCC)	Inappropriate alternative text (p. 12) Poorly organized content sequence (p. 12) Inappropriate link text (p. 13) Illogical order of focus (p. 13) Insufficient labels (p. 13)
National Health Insurance Service (NHIS)	Inappropriate alternative text (p. 13) Poorly organized content sequence (p. 14) Unexpected response to user request (p. 14) Insufficient labels (p. 14) Inappropriate link text (p. 14) Inappropriate hidden message (p. 15)
National Health Information Portal (NHIP)	Inappropriate page title (p. 15) Poorly organized content sequence (p. 15) Inappropriate link text (p. 15) Unnecessary glossary links hampering ease of reading (p. 16) Inappropriate alternative text (p. 16)
Ministry of Health and Welfare (MOHW)	Inappropriate alternative text (p. 16) Poorly organized content sequence (p. 16) Inappropriate tab title (p. 16) Inappropriate link text (p. 17) Insufficient labels (p. 17) Problems with the accessibility of Web applications (p. 17)
Korea Centers for Disease Control and Prevention (KCDC)	Inappropriate alternative text (p. 17, p. 18) Incorrect markup (p. 17) Inappropriate page title (p. 18) Illogical order of focus (p. 18)
National Medical Center (NMC)	Inappropriate alternative text (p. 18) Lack of mechanism to bypass blocks (p. 18) Illogical order of focus (p. 19) Poorly organized content sequence (p. 19) Insufficient labels (p. 19)
National Rehabilitation Center (NRC)	Poorly organized content sequence (p. 19) Difficult to comprehend tables (p. 19) Compatibility problems between keyboard operation and screen reader (p. 20) Problems with the accessibility of Web applications (p. 20)
Public Health Portal (G-health)	Compatibility problems between keyboard operation and screen reader (p. 20) Unexpected response to users' request (p. 20) Lack of mechanism to bypass blocks (p. 20) Inappropriate alternative text (p. 21) Poorly organized content sequence (p. 21)
Ministry of Food and Drug Safety (MFDS)	Poorly organized content sequence (p. 21) Inappropriate alternative text (p. 21) Unclear instructions (p. 21)
National Center for Mental Health (NCMH)	Inappropriate alternative text (p. 22) Illogical order of focus (p. 22) Insufficient labels (p. 22) Inappropriate alternative text (p. 22) Incorrect markup (p. 22)

While performing the first task, users indicated the lack of appropriate alternative text. For example, when the screen reader read 'select', users understood that they had already selected one, though they actually had not until they pressed 'enter' again, which left users feeling confused.

In the second task, page navigation, none of the users failed. It took them about 1 and a half to 3 min, 2 min on average to complete the task. Users highlighted content

sequence problems. The hierarchical structure of the main menu was difficult for them to understand. Even though 'Health Insurance System Overview' is a sub-menu of 'National Health Insurance', it is read in the same hierarchy as if they were both part of the main menu, which hindered users navigating the Web site and locating information they seek as presented in Fig. 2. None failed in the third task, search function. It took about 1–4 and a half minutes, 2 min

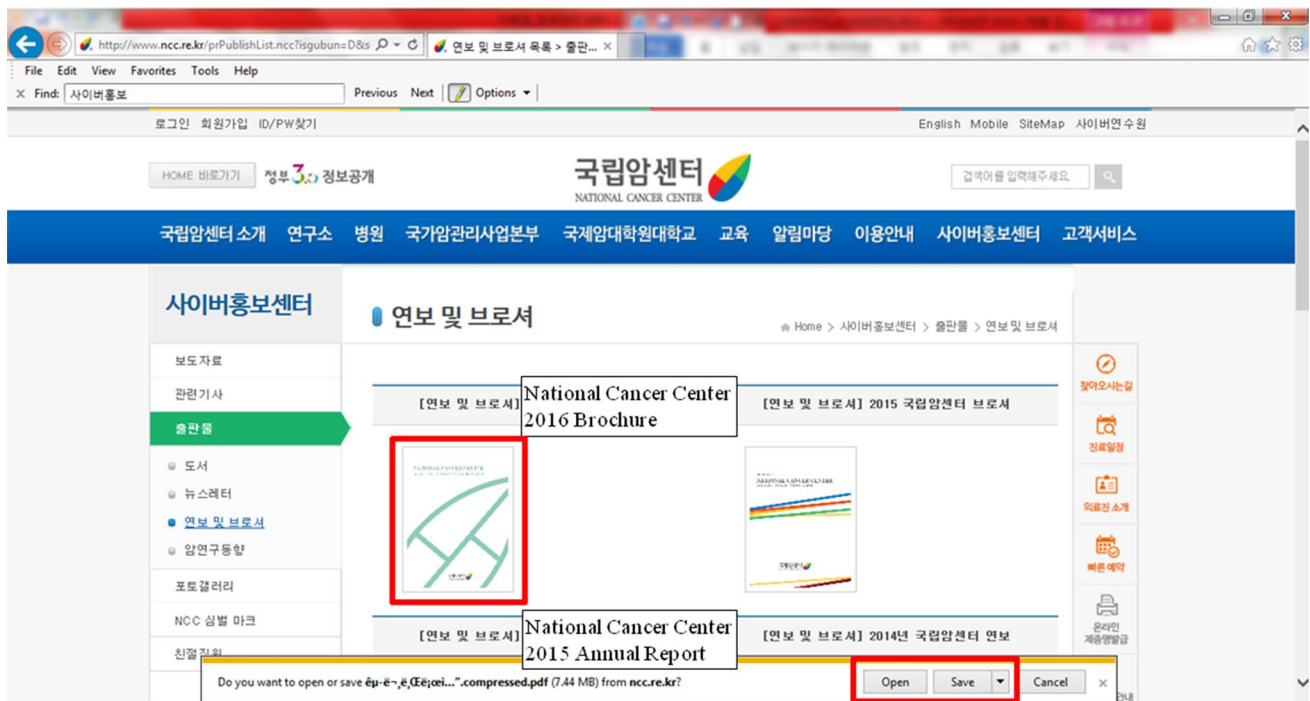


Fig. 1 A link text error in NCC



Fig. 2 A content sequence error in NHIS

on average to complete the task. There were inappropriate responses for user requests. For example, when a user moved the focus out of the search box using the direction key to

check whether the query was written correctly, the search term was erased unexpectedly; therefore, the user had to rewrite the search term, which left users feeling frustrated.

While performing download, one user failed. It took about 1 and a half to 6 min, with 3 min on average to complete the task. Users addressed label problems and the lack of appropriate link text. For example, although there is a checkbox to input ‘Resident Registration Number’, it cannot be identified because the checkbox is not labeled. In addition, the Web site does not provide a link text indicating that the link of the ‘Issue print’ button will pop up in a new window. Regarding the last task, form creation, seven users failed. It took them about 8–16 min, 10 min on average to complete the task. Users highlighted incompatibility with screen readers and the lack of alternative text. When verifying user identity through a mobile phone, there was no sound for the security codes. Moreover, it is impossible to proceed to the next step because there is no alternative text for the condition, the ‘Under 14 years old’ or ‘Over 14 years old’. Due to inappropriate hidden text, the screen reader read the phrase ‘Enter the second digit of your resident registration number’, although actually, there was nothing on the screen.

Third, the study tested the National Health Information Portal (NHIP). NHIP delivers reliable, comprehensive, and integrated online health information provided by health professionals [30]. None failed in registering on the NHIP site. It took about 1–3 min, 2 min on average, to complete the task. Users found that the Web pages did not provide titles that described appropriate topics. The title of the main page read ‘Enter your information and complete sign up’. This title confused users regarding whether they were currently

in the process of entering information or whether the sign-up was completed. For the second task, page navigation, none failed. It took them from 1 to 10 min, 4 min on average, to complete the task. The directory of diseases was tricky for users. The Web page listed the diseases under the categories in alphabetic order; if a user wanted to find information about pneumonia, he or she needed to go through all the diseases from A to O and move to P categories until finally reaching to pneumonia. Accordingly, users commented that without the categories, it would be easier to search a specific disease because the structure of this site was not easy for a visually impaired person to figure out. In addition, users found proper link text lacking. The site had links titled, for instance, ‘Gastric cancer’ and ‘Pituitary gland’, however they did not understand what these were as the site did not clarify that these are popular search terms. One of the notable problems on the NHIP site is that there were too many glossary links for even unnecessary words, which interrupted a user’s understanding of the whole content as shown in Fig. 3.

Regarding the form creation task, users reported inappropriate page titles. The post list page and the writing page were titled exactly the same, as ‘the post’; thus, users were confused which page they were currently on.

Fourth, the study tested the Ministry of Health and Welfare (MOHW) Web site. MOHW provides the public with easy access to policy and administrative information on health and welfare [31]. None failed in page navigation. It

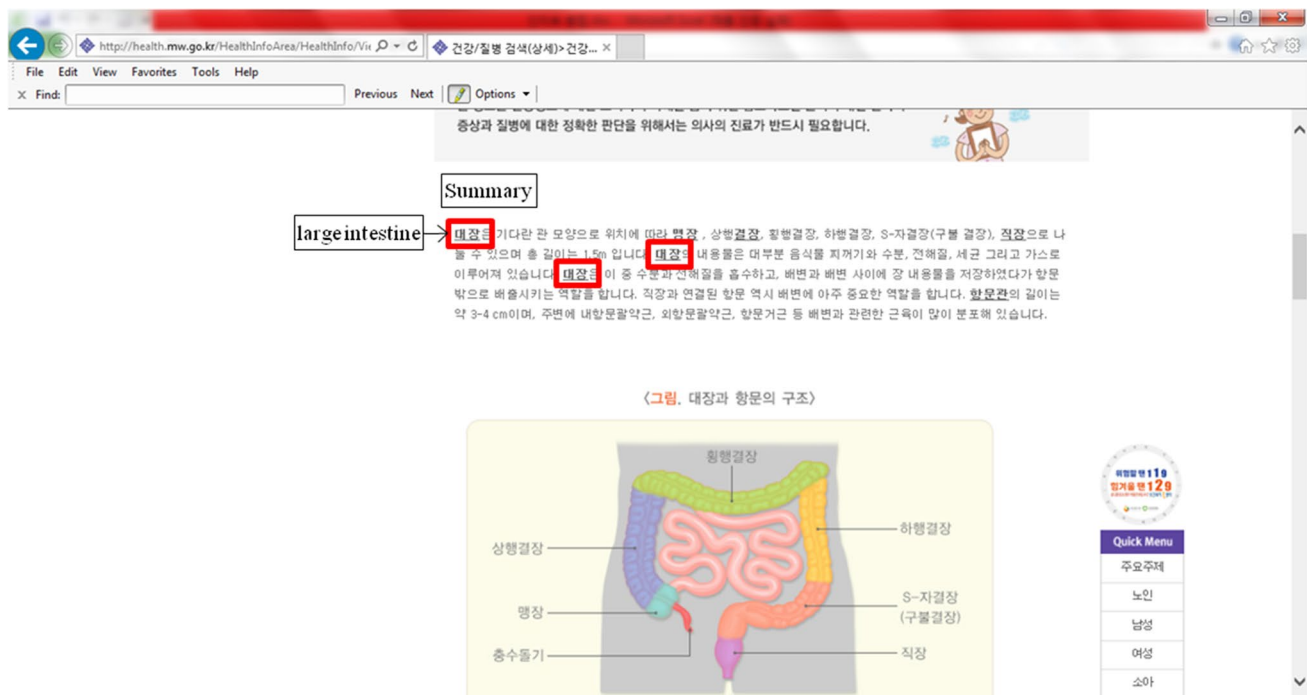


Fig. 3 Too many glossary links in NHIP

took about 2–8 min, 4 min on average to complete the task. The main page did not provide any information on the hierarchical structure between the main menus and sub-menus, which resulted in content sequence problems. The search function task took about 1–7 min, 4 min on average, to complete. Users reported tab title errors on the search results page. Tab titles such as ‘Title1 link’ and ‘Title2 link’ did not explain how many tabs there were as shown in Fig. 4. Thus, users recommended providing more specific information about the tab title, such as ‘Title1 Table 1/3 link’ and ‘Selected Title2 Table 2/3 link’ to indicate the tab structure.

While performing the fourth task, download, none failed. It took about 30 s to 3 and a half minutes, 2 min on average, to complete the task. Users highlighted the lack of link text, because of which they could not recognize that files were attached. Thus, users recommended that the site provide the appropriate link text (e.g., file attached) or a download button right beside an attached file. None failed in form creation. It took about 4 and a half to 10 min, 7 min on average, to complete the task. Users reported two main issues. One was about inappropriate labels. There was no label for the date of birth format. Users were confused about whether they should input 8 (yyyy/mm/dd) or 6 digits (yy/mm/dd). Thus, they often needed to re-input their date of birth. The other was a Web application problem. CAPTCHA, a Web application regarding security, was difficult to listen to and was even pronounced in a foreign language, e.g., English,

rather than Korean, thus, users had trouble when they tried to input the security characters, as shown in Fig. 5.

Fifth, the study tested the Korea Centers for Disease Control and Prevention (KCDC) Web site. The KCDC addresses prevention and control, quarantine, and research for infectious diseases, chronic diseases, and rare diseases [32]. None of the users failed in page navigation. It took them 30 s to 7 min, 3 min on average, to complete the task. Users indicated issues of alternative text and markup correctness. As presented in Fig. 6, no alternative text was provided for the images of ‘Domestic/overseas outbreak status’; they were identified as just ‘image’. Thus, there is no way to understand what the image presents. For markup correctness problems, users found that even though there was a file under ‘relevant information’, the tag for ‘list start count’ read 0 as if there was no file.

Regarding the search function task, none of the users failed. It took them about 30 s to 3 min, 1 min on average, to complete the task. Users found three major errors. First, the title of the search results page was the same as the site name,



Fig. 5 A web application error in MOHW

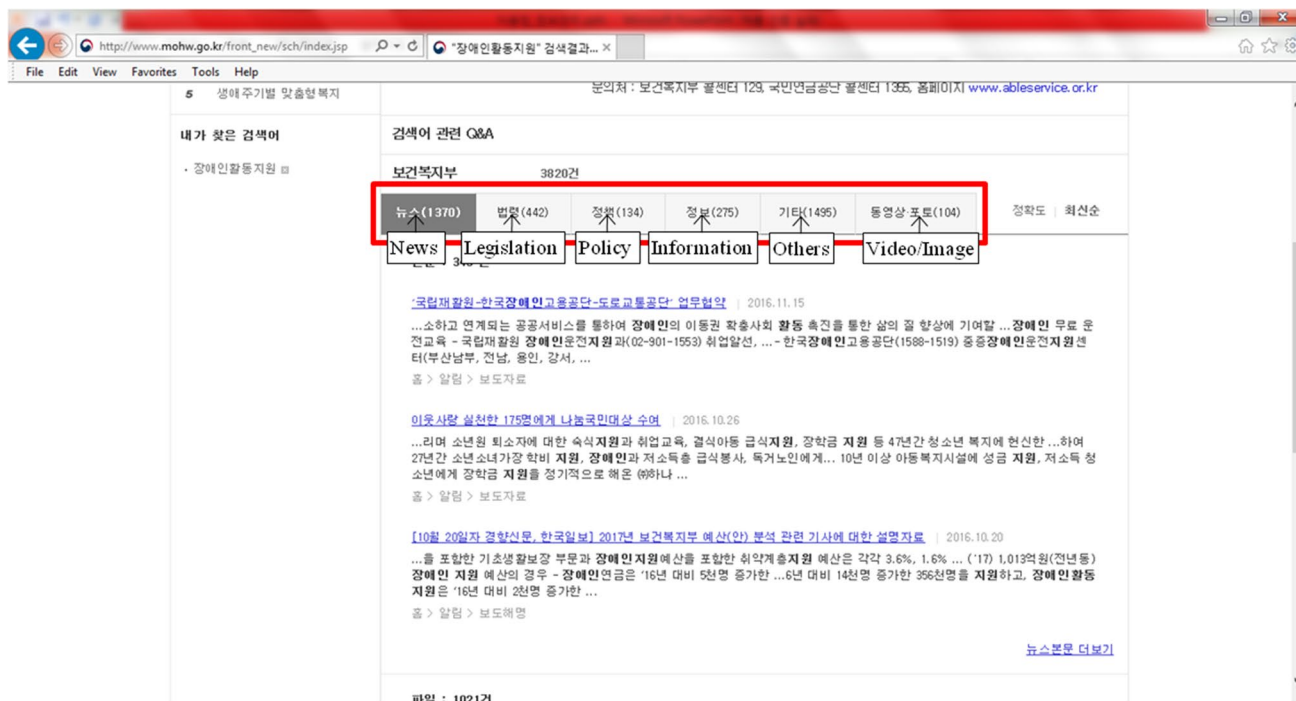


Fig. 4 A tab title error in MOHW



Fig. 6 An alternative text error in KCDC

thus, users were confused about whether they were still on the main page or in the search results page. Second, there was a focus order problem. When setting a search period for the advanced search, the focus should move to the calendar table as soon as the ‘calendar’ link is selected, however it did not. Third, when setting a search period, no specific alternative text was called start date/end date.

Sixth, the study tested the National Medical Center (NMC) Web site. NMC provides healthcare services for the public as a network hub of nationwide public healthcare institutions and hospitals [33]. Regarding page navigation, none of the users failed. It took them about 1–7 min, 4 min on average, to complete the task. Users addressed issues of alternative text and bypass blocks. For a bypass blocks error, the site did not have sub-menu or main menu shortcuts; thus, users could not skip repeated areas of content. None failed in the download tasks. It took the users about 1–5 min, 2 min on average to complete the task. Users reported that download links were not identified as “download”. Rather, it just read ‘hwp’ (a Korean file format). Regarding those issues, experts pointed out that when a non-expert user hit an enter button on the download link, the user expected a new window to open; however, unexpectedly, when only the content changed on the same Web page, they would have confused, thus, the link had to be opened in a new window.

While performing the form creation task, none of the users failed. It took them about 3–18 and a half minutes, 13 min on average, to complete the task. Users identified

several problems. Focus order errors were severe. Users could not be sure whether the medical department, medical staff, or appointment time were selected. If one of them were selected, the site should tell the user, and the focus should move to the next combo box, though it did not. As for content sequence errors, users had a hard time grasping how to use ‘Mobile phone verification (by) App’ and ‘Mobile phone verification by text messages’ links, as shown Fig. 7. Thus, users recommended that these choices should be provided through a checkbox or combo box rather than links.

Seventh, the study tested the National Rehabilitation Center (NRC) Web site. The NRC is dedicated to health promotion and community reintegration for people with disabilities through rehabilitation medicine, research and education [34]. Page navigation took about 1–5 min, 3 min on average, to complete. Users noted issues of content sequence and tables. There were too many top menus on the main page (75 menus), not organized appropriately. Experts commented that considerable time was required to navigate such Web sites without using the skip navigation function; however, most of the non-experts did not even know such a function existed. Above all, the ‘medical appointment schedule’ was provided in a table format, which was difficult for persons with visual impairment to understand, as presented in Fig. 8. When there is no appointment scheduled, the dates in the table were blank. Users were confused with all the empty cells, because there was no alternative text for them;



Fig. 7 A content sequence error in NMC

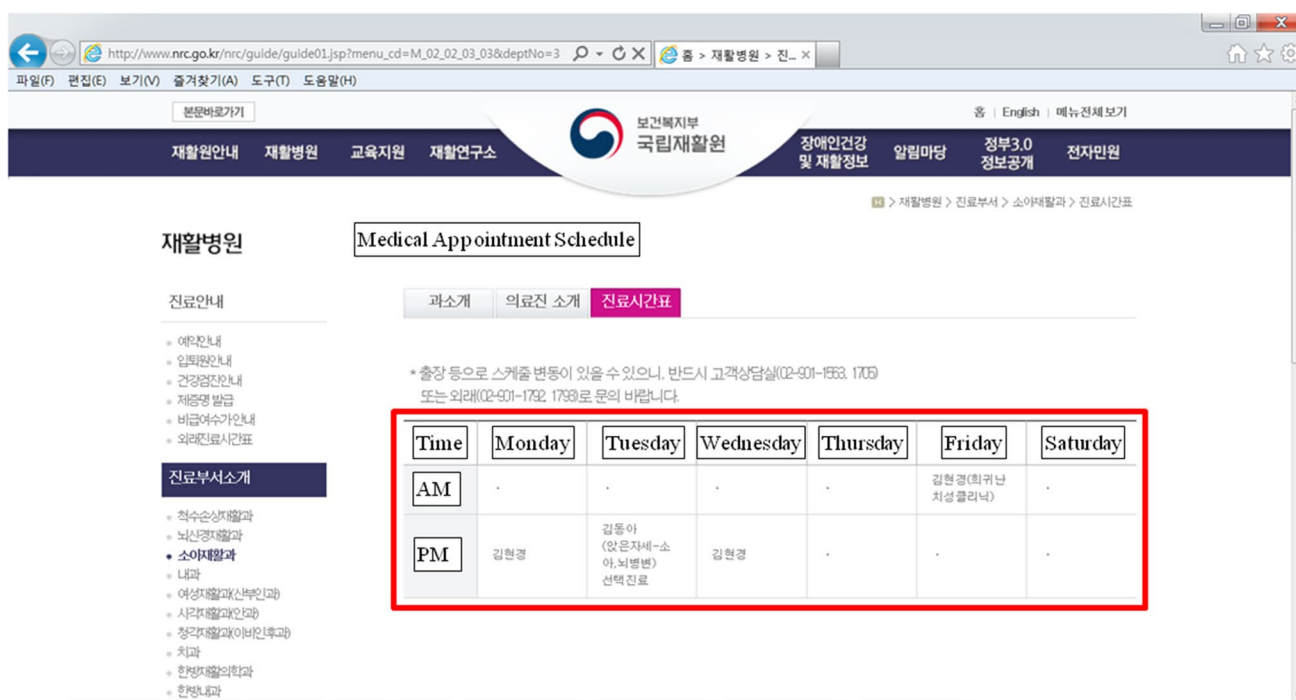


Fig. 8 A table error in NRC

thus, they perceived that an appointment was not available on those dates.

In the third task, search function, one user failed. It took about 1–2 min, 1 and a half minutes on average, to complete the task. Users reported content sequence problems. While the search box is placed on the top in general, it was unexpectedly located at the bottom of the page, which made users uncomfortable; they needed to scroll all the way down to the bottom of the page to use the search box. In the form creation task, one user failed. It took about 4–19 min, 11 and a half minutes on average, to complete the task. Users found keyboard operation errors. When entering the user name, phone number, and birth date in the text entry box, the keyboard is not read, that is, a screen reader did not recognize them.

Eighth, the study tested the Public Health Portal (PHP) Web site. PHP provides customer-oriented healthcare and related services through the utilization of the regional healthcare information systems [35]. While registering on PHP, the screen reader could not read what users typed and deleted. Without failure, page navigation took about 6–17 min, 10 min on average. In this task, two main issues were observed. One was user response errors. Tabs were automatically switched whenever the focus moved, which was abnormal. That is, tabs are to be switched when a user clicks on enter or clicks to switch. The other was a bypass blocks problem: a user can select the ‘search’ button only after going through the entire ‘medical department’ list even

though he/she may have already chosen one. None of the users failed in form creation. It took them about 3–9 min, 4 min on average to complete the task. There were content sequence and alternative text problems. Regarding content sequence problems, selecting a local healthcare organization was complicated because the hierarchical structure of menu options was not logical. Users found incorrect alternative texts; a combo box was recognized as “a link”.

Ninth, the study tested the Ministry of Food and Drug Safety (MFDS) Web site. MFDS promotes the public health by ensuring the safety of all foods, drugs, cosmetics, herbal medicines, and medical devices [36]. When performing page navigation, none of the users failed; it took them about 1–5 min, 3 min on average, to complete the task. Users encountered content sequence problems. As shown in Fig. 9, the structure of menus was not consistent; the submenu structure under the main menu was different from the other menus, which confused the users. When users went into the ‘Food safety’ menu, an irrelevant page suddenly appeared, entitled ‘Food administrative disposition’, which left users frustrated; they were not sure about whether they were on the right page or not.

None of the users failed in the search function. It took them about 1–3 min, 1 and a half minutes on average, to complete the task. Users were confused when they found the same alternative text for the main search box and the bulletin board search box. In the download tasks, no user failed; it took them about half to 4 min, 2 min on average, to complete

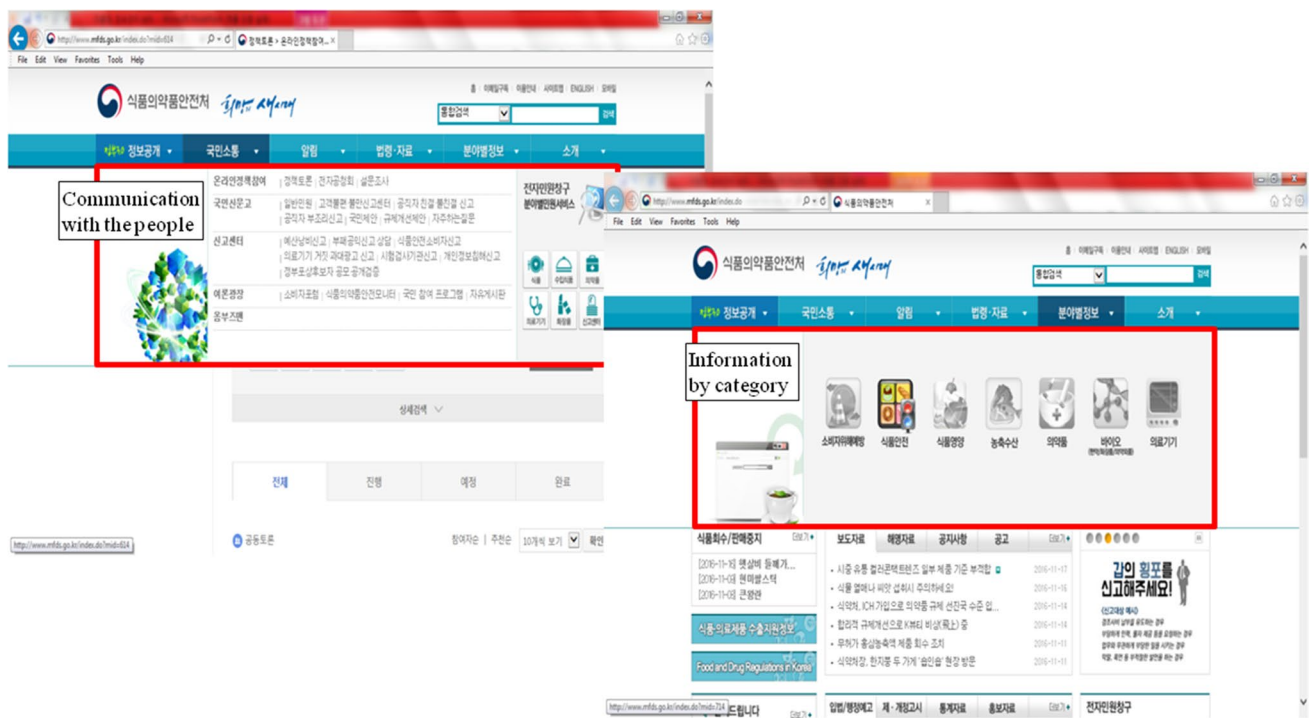


Fig. 9 A content sequence error in MFDS

the task. It was difficult to recognize the item as downloadable, because it just said the title and the file format (e.g., pdf) at the end. Thus, users suggested that if it was a file for downloading or saving, providing saving/downloading buttons would be more recognizable.

Finally, the study tested the National Center for Mental Health (NCMH) Web site. NCMH pursues new systems for public mental health, and supports evidence-based research for mental health improvement [37]. None of the users failed in the form creation task. It took them about 6–19 and a half minutes, 11 min on average, to complete the task. However, users identified several problems. First, there were focus order problems. As a user went through steps 1, 2, and 3 on the appointment page, the cursor should logically move to the next step after selecting one step, rather than moving back to the very top of the page; due to that error, a user had to repeat unnecessary movement. Next, users reported markup correctness problems. The ‘Select appointment’ list box read ‘three’, though there are only “two” options for morning and afternoon. Lastly, the page provided the unclear alternative text “Select link”, rather than specifically describing “Select appointment date” or “Select appointment doctor”, as presented in Fig. 10.

5.2 Discussion

Considering the problems identified and the time taken to perform the tasks, the National Medical Center (NMC),

National Rehabilitation Center (NRC) and the National Health Insurance Service (NHIS) Web sites were the least accessible, whereas the Ministry of Food and Drug Safety (MFDS) and the Korea Centers for Disease Control and Prevention (KCDC) Web sites were the most accessible. The evaluation results regarding the MFDS and the KCDC Web sites were consistent with the results of previous automated testing, while the evaluation results concerning the NMC and NRC were significantly different from those of the automated testing [9]. In the previous automated testing, the NMC and NRC Web sites attained perfect scores, whereas task incompleteness occurred only in the two Web sites in the present study. Some findings of the present study overlapped with the previous testing [9]. The key issues were the lack of alternative text and link text, incompatibility with screen readers, bypass blocks problems, content sequence errors, poor table composition, keyboard operation errors, and markup errors.

However, the results of user testing revealed the accessibility issues in detail, which the two previous tests did not find; among the problems identified, study participants regarded Web application errors as one of the biggest obstacles. Web application errors occurred in the user verification process, which caused challenging issues for registration or form creation. Users cannot register a healthcare Web site or make an appointment with healthcare professionals until completing that process. Due to that process, completing the registration or creating the form took the longest time among

The screenshot shows the NCMH Web site interface. A red box highlights the 'Enter appointment information' section, which contains a table with fields for Date, Time, and Doctor. The table has a 'Select' button next to the Time field.

Enter appointment information	
Date	예약일자 선택
Time	예약시간 선택
Doctor	예약의사 선택

The 'Select' button is located to the right of the 'Time' field.

Fig. 10 An alternative text error in NCMH

the tasks imposed; users spent approximately 3–26 min and 11 min, respectively, on average on these tasks, and some users failed in registration, download, or form creation. These obstacles not only hampered accessibility but also left users feeling frustrated. Compared with healthcare Web sites in the US [18] and general Web sites, the healthcare Web sites in Korea are unique in that the majority of Web sites require users to register or log in through user verification processes because some tasks (e.g., medical appointments, health insurance, etc.) collect and secure personal data. Otherwise, users cannot fully access the information that they provide or users cannot comment on the bulletin board or post questions there. The user verification process involves Web application errors, which were not identified in previous studies regarding the accessibility of government Web sites [16, 17].

In the past, users were primarily informed from the healthcare Web sites by browsing or searching the Web sites; whereas recently, users have obtained more services by downloading relevant files from Web sites or by asking questions on Web sites to meet their individual information needs. Therefore, the findings of the present study highlight the importance of Web accessibility as fundamental for facilitating interactions between Web sites and users without discrimination. Fortunately, it is not difficult to address most of the problems identified in the present study. First of all, the healthcare Web sites can fix the critical problems observed according to the specific solutions that accessibility experts who participated in the study suggest, as stated in the findings. Moreover, the lack of appropriate alternative text, link texts, labels, or titles—the challenging issues which are most frequently reported—can be fixed by simply providing appropriate descriptions with text in the designated places. The other issues, such as incompatibility with screen readers and keyboards and problems of bypass blocks, focus order, and markup correctness, can be fixed by editing the source codes. Web application errors such as CAPTCHA, were identified as one of the biggest challenges, which, should thus be replaced by accessible alternative applications. With simple fixes, most of the Web sites in the study can attain greater accessibility. To develop accessible Healthcare Web sites, above all, it is essential that the government and public agencies have further funding to support various testing and procurement of trained Webmasters who have sufficient knowledge of KWCA 2.1.

6 Conclusions

Healthcare Web sites of Korean government and public institutions have revealed issues and problems across the four dimensions of accessibility, perceivable, operable, understandable, and robust. First, according to principle

1, perceivable, the main errors occurred in providing adequate alternative text and clear instructions. Second, five errors were predominant regarding principle 2, operable. These were keyboard operation errors, focus order errors, bypass blocks errors, page title errors, and the lack of appropriate link text. Third, with respect to principle 3, understandable, most problems were identified with regard to user responses, content sequence, tables, and labels. Finally, based on principle 4, robust, problems regarded markup correctness and Web application accessibility. As noted in previous studies [16, 18], the problems identified can be simply resolved; the experts who participated in the present study made the same suggestions. However, due to the lack of policies or regulations that require compliance with Web accessibility standards or due to government or public agencies' lack of awareness for accessibility [9], similar accessibility issues have recurred. To understand the main factors that affect compliance with Web accessibility guidelines, perceptions of the government agencies or their Web developers must be evaluated [17]. Therefore, future research needs to address whether government and public agencies or their Web developers are aware of the importance of accessibility.

The findings and suggestions of this study can provide helpful references for accessible Web site design in other contexts. When accessing the Internet through smartphones, users access mobile Web sites usually built in a PC environment. Thus, similar accessibility problems appear to occur on mobile devices as well. Considering the prevalence of smartphones, the findings of the present study can be applied to developing mobile services of healthcare and allied organizations for people with visual impairment, thereby reducing the digital information gap resulting from accessibility issues. Since smartphones play a key role in information seeking, future research needs to pay attention to mobile accessibility issues and related policies. As public libraries develop a lot of mobile Web sites [38], it is necessary to investigate mobile accessibility to public libraries. Improving accessibility of healthcare Web sites of government and public institutions not only helps the elderly and people with vision impairment to use reliable health information quickly and comfortably, but also contributes to the dissemination of public health information, thereby substantially promoting public health. The effort to attain Web accessibility of government/public agencies can be a desirable approach to achieve the goals of Government 3.0, which seeks to share information with members of the public and to promote communication with them.

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